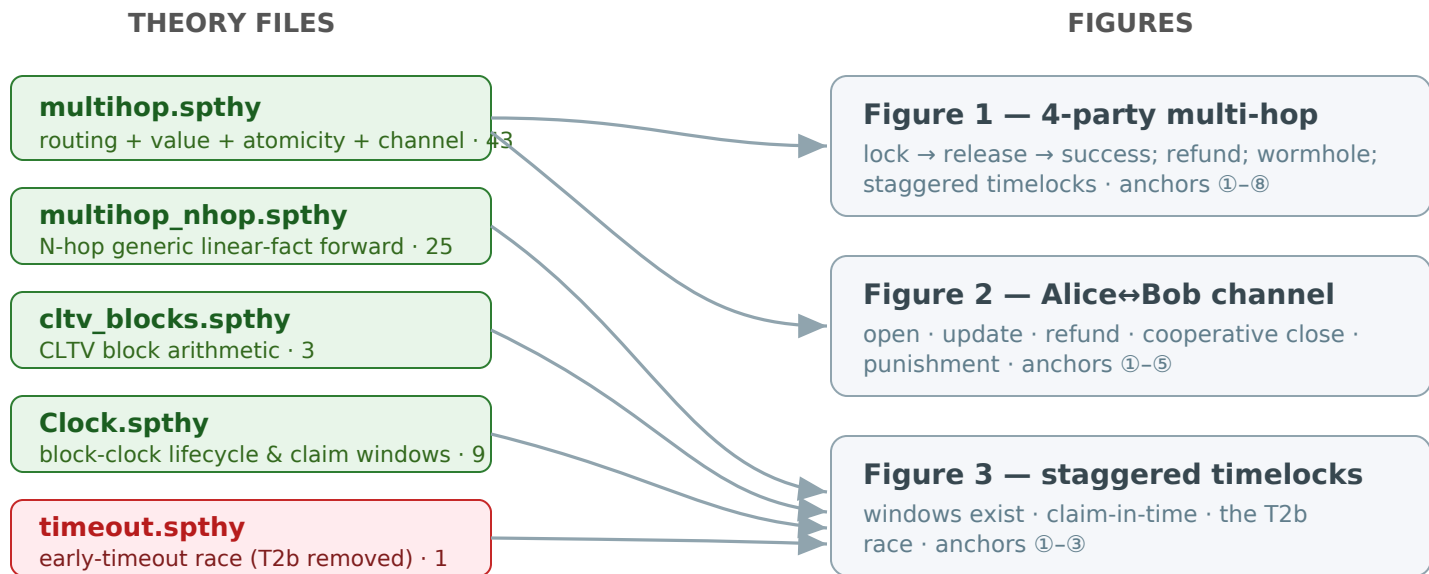


Verification map — 5 theories · 4 figures · 81 lemmas

where each file's lemmas live, and how the figures fit together



COLOUR KEY

- honest success / safety proved
- benign failure (timeout / refund)
- malicious / attack (reachable)

TOTALS

multihop 43 · nhop 25 · Clock 9 · cltv 3 · timeout 1

81 / 81 machine-checked ✓

(Tamarin 1.8, per-lemma to avoid OOM)

Read order: Fig 2 (one channel) →
Fig 1 (route over channels) →
Fig 3 (why the timing is safe).

multihop_nhop.spthy is the N-hop generalisation (25 lemmas, ~103.5s); the reusable-channel boundary spike is documented separately. See nhop_soundness for the refinement theorem transferring safety to all N.